

Solving Absolute Value Equations

Algebra Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Understand that absolute value represents distance from zero and is always non-negative
- Solve absolute value equations by splitting into two separate equations (positive and negative cases)
- Isolate the absolute value expression before splitting into two equations when coefficients or constants are present

Problems

1. Solve the absolute value equation for x.

$$|x| = 9$$

2. Solve the absolute value equation for x.

$$|x + 3| = 7$$

3. Solve the absolute value equation for x.

$$|x - 6| = 4$$

4. Solve the absolute value equation for x. Check whether a solution exists.

$$|x + 2| = -5$$

5. Solve the absolute value equation for x.

$$3|x| = 18$$

6. Solve the absolute value equation for x.

$$2|x + 3| = 10$$



7. Solve the absolute value equation for x.

$$4|x - 1| - 8 = 12$$

8. Solve the absolute value equation for x.

$$3 + 5|x - 2| = 13$$

9. Solve the absolute value equation for x. Then verify both solutions by substituting them back into the original equation.

$$\frac{1}{2}|2x + 4| - 3 = 5$$

10. Solve the absolute value equation for x. Verify both solutions by substituting them back into the original equation.

$$2|3x - 1| + 7 = 19$$



Solving Absolute Value Equations — Answer Key

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Answer Key

1. Answer: $x = 9$ or $x = -9$

- Split into two equations: $x = 9$ and $x = -9$
- Both values satisfy the original equation since $|9| = 9$ and $|-9| = 9$

2. Answer: $x = 4$ or $x = -10$

- Split into two equations: $x + 3 = 7$ and $x + 3 = -7$
- Solve first: $x = 7 - 3 = 4$
- Solve second: $x = -7 - 3 = -10$

3. Answer: $x = 10$ or $x = 2$

- Split into two equations: $x - 6 = 4$ and $x - 6 = -4$
- Solve first: $x = 4 + 6 = 10$
- Solve second: $x = -4 + 6 = 2$

4. Answer: No solution

- Absolute value is always non-negative (it represents distance from zero)
- An absolute value expression can never equal a negative number
- Therefore, there is no solution

5. Answer: $x = 6$ or $x = -6$

- Isolate the absolute value by dividing both sides by 3: $|x| = 6$
- Split into two equations: $x = 6$ and $x = -6$

6. Answer: $x = 2$ or $x = -8$

- Isolate the absolute value by dividing both sides by 2: $|x + 3| = 5$
- Split into two equations: $x + 3 = 5$ and $x + 3 = -5$
- Solve first: $x = 5 - 3 = 2$
- Solve second: $x = -5 - 3 = -8$

7. Answer: $x = 6$ or $x = -4$

- Add 8 to both sides: $4|x - 1| = 20$
- Divide both sides by 4: $|x - 1| = 5$
- Split into two equations: $x - 1 = 5$ and $x - 1 = -5$
- Solve first: $x = 5 + 1 = 6$
- Solve second: $x = -5 + 1 = -4$

8. Answer: $x = 4$ or $x = 0$

- Subtract 3 from both sides: $5|x - 2| = 10$
- Divide both sides by 5: $|x - 2| = 2$





- Split into two equations: $x - 2 = 2$ and $x - 2 = -2$
- Solve first: $x = 2 + 2 = 4$
- Solve second: $x = -2 + 2 = 0$

9. Answer: $x = 6$ or $x = -10$

- Add 3 to both sides: $\frac{1}{2}|2x + 4| = 8$
- Multiply both sides by 2: $|2x + 4| = 16$
- Split into two equations: $2x + 4 = 16$ and $2x + 4 = -16$
- Solve first: $2x = 12$, so $x = 6$
- Solve second: $2x = -20$, so $x = -10$
- Verify $x = 6$: $\frac{1}{2}|12 + 4| - 3 = \frac{1}{2}(16) - 3 = 8 - 3 = 5$ ✓
- Verify $x = -10$: $\frac{1}{2}|-20 + 4| - 3 = \frac{1}{2}(16) - 3 = 8 - 3 = 5$ ✓

10. Answer: $x = 7/3$ or $x = -5/3$

- Subtract 7 from both sides: $2|3x - 1| = 12$
- Divide both sides by 2: $|3x - 1| = 6$
- Split into two equations: $3x - 1 = 6$ and $3x - 1 = -6$
- Solve first: $3x = 7$, so $x = 7/3$
- Solve second: $3x = -5$, so $x = -5/3$
- Verify $x = 7/3$: $|3(7/3) - 1| = |7 - 1| = |6| = 6$, then $2(6) + 7 = 19$ ✓
- Verify $x = -5/3$: $|3(-5/3) - 1| = |-5 - 1| = |-6| = 6$, then $2(6) + 7 = 19$ ✓

